

Shanu Kumar

Platform Engineering | AWS & Kubernetes | Internal Developer Platforms

shanukumarm@gmail.com | +91-7631340415 | Hyderabad, India | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Platform Engineering Lead with 5 years building Internal Developer Platforms on AWS and Kubernetes, focused on self-service infrastructure, developer experience, and platform reliability. Designed and scaled IDP infrastructure for 20+ engineering teams using Terraform CDK construct libraries, Istio ambient mesh, and KEDA-driven CI runner fleets, achieving 90% provisioning effort reduction, 25% mesh overhead reduction, and 30% CI cost reduction.

EXPERIENCE

Lead Engineer

Verisk Analytics

Apr 2024 – Present

Hyderabad, India

- **Platform architecture & scale:** Standardised infrastructure onboarding for 20+ teams across 500+ daily pipeline runs by architecting a per-team isolated AWS account model with automated bootstrapping and language-specific CI/CD templates.
- **Infrastructure automation:** Reduced provisioning effort by 90% with CDKTF TypeScript constructs using a main-service + supporting-services pattern; a pre-flight registration check enforces mandatory billing tags, preventing configuration drift across the platform.
- **Kubernetes & service mesh:** Reduced service mesh resource consumption by 25% with zero downtime by migrating Istio sidecar to ambient mode across 20+ EKS clusters; migrated 100+ services from Istio Ingress to Kubernetes Gateway API, separating infrastructure and application routing ownership.
- **CI/CD platform & security:** Reduced CI costs by 30% by replacing Azure-hosted agents with self-hosted EKS runners scaled by KEDA and Karpenter; integrated DependencyTrack for supply chain vulnerability scanning across 100+ services.
- **AI platform assistant:** Built an internal RAG assistant on AWS Bedrock for natural-language access to platform documentation for 20+ teams; secured with Okta JWT, Kong API Gateway rate limiting, and Bedrock Guardrails.

Technical Lead

HCL Technologies

Oct 2022 – Mar 2024

Pune, India

Promoted from Senior Software Engineer

- **Shared EKS platform:** Reduced infrastructure configuration effort by 85% by building a multi-tenant EKS platform for 30+ teams with namespace isolation, per-team Karpenter node pools, and reusable Terraform and Helm templates.
- **Multi-account networking:** Eliminated manual routing changes on new team onboarding by redesigning Transit Gateway from flat hub-and-spoke to a hierarchical model using RFC 6598 supernets, scaling to 30+ teams without CIDR exhaustion or on-prem address conflicts.
- **CI/CD standardisation:** Cut release cycle time by 50% for 10+ teams by delivering a shared CI/CD pipeline with SBOM generation, container scanning, and ArgoCD GitOps; reduced time-to-production for every new team onboarded.
- **AWS account governance:** Reduced shared AWS account costs by 60% by enforcing least-privilege IAM with mandatory tagging policies and automating resource cleanup via EventBridge-triggered Lambda jobs; governed 20+ engineers with zero untagged resource drift.

Senior Software Engineer

HCL Technologies

Aug 2021 – Oct 2022

Pune, India

- **Kubernetes platform bootstrap:** Cut Kubernetes onboarding time from 2–3 days to 4 hours for 10+ teams by building a reusable Helm chart library covering the full application deployment lifecycle.
- **Deployment strategy & GitOps:** Standardised safe delivery across 50+ services by introducing blue-green and canary deployment strategies with ArgoCD GitOps, eliminating manual release steps.
- **IaC foundations:** Established the org's Terraform module library, remote state foundation, and GitLab CI plan/approval gate from scratch; adopted as the infrastructure delivery standard across all teams.

TECHNICAL SKILLS

- **Infrastructure & Cloud:** Terraform, CDKTF, CloudFormation, AWS (EKS, EC2, RDS, S3, VPC, IAM, Lambda, DynamoDB, SNS/SQS, Bedrock, Route 53, CloudFront, EventBridge, API Gateway, OpenSearch)
- **Containers & Orchestration:** Kubernetes, Docker, Helm, KEDA, Karpenter, Istio (sidecar & ambient), Gateway API, cert-manager, Harbor, Rancher
- **CI/CD & GitOps:** Azure DevOps, GitHub Actions, GitLab CI, Jenkins, ArgoCD, JFrog Artifactory
- **Scripting & Automation:** Python (FastAPI, boto3), TypeScript, Bash, PowerShell
- **Security & Observability:** Trivy, SonarQube, DependencyTrack, Bedrock Guardrails, Okta IAM, Kong, Datadog, PagerDuty, Prometheus, Loki, Grafana

CERTIFICATIONS

- **Certified Kubernetes Administrator (CKA)**, CNCF, 2023
- **Certified Kubernetes Application Developer (CKAD)**, CNCF, 2023
- **HashiCorp Terraform Associate (003)**, HashiCorp, 2023
- **AWS Certified AI Practitioner**, AWS, 2026

EDUCATION

Master of Technology in Computer Science <i>Indian Institute of Technology Madras</i>	Jul 2019 – Jul 2021 <i>Chennai, India</i>
Bachelor of Technology in Computer Science <i>Loknayak Jai Prakash Institute of Technology</i>	Sep 2014 – Jul 2017 <i>Chapra, India</i>